Fraud & Anomaly Detection

2026-04-01 to 2026-09-30 · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0009
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Executive summary

Demonstration report. Built entirely from synthetic data generated by ProQuant for illustration. The company, the customers, the merchants and the transactions are fictitious, and far more fraud is planted than a real book would hold, so that each pattern can be seen.

ProQuant Analytics reviewed 82,192 debit transactions of Demo Client (2026-04-01 to 2026-09-30) to answer three questions: what do the current controls really catch, what are they missing, and what should the team do with the capacity it has? Behavioural features were built with no information from the future; transparent rules, an anomaly model and a supervised model were combined into one score; the score was tested on a period it had never seen, at the number of alerts the team can review; and the alerts were grouped into cases for the investigators. Nothing here judges any person: it shows where to look next.

Overall view: Fraud controls have material gaps

11 of 25 tests passed, 11 with a warning, 3 failed. Conclusions are drafted from the figures and reviewed by the analyst before release.

TRANSACTIONS SCORED	FRAUD VALUE FOUND	PRECISION (LOWER BOUND)
82,192	**99%**	**16.2%**
34.54m USD · 466 confirmed	current rules 57%	current rules 4.7%

CURRENT ALERTS A DAY	CASES TO REVIEW	NET GAIN A YEAR
14.0	**346**	**417k**
capacity 12; 16% aged	8 P1 · 1.02m USD unconfirmed	USD, on confirmed fraud alone

Conclusions

- **Headline** 82,192 debit transactions worth USD 34.54m (2026-04-01 to 2026-09-30) were scored; 466 are confirmed fraud worth USD 804k (232.6 bps of the value). The combined score was tested on 2026-08-21 to 2026-09-15, a period the models never saw.
- **Detection** At the threshold set for 12 alerts a day the score produced 18.8 a day, found 99% of the confirmed fraud value and 93% of the cases, with a confirmed precision of 16.2% (interval 12.8% to 19.7%). The current rules: 14.4 alerts a day, 57% of the value, 21% of the cases, precision 4.7%.
- **Labels** Precision here is a lower bound: a transaction counts as fraud only if someone confirmed it, and nobody has looked at most of what the score raises. The comparison favours the current rules, because part of the confirmed cases exist only because those rules fired. Half of the cases are confirmed within 12 days, so the last 30 days are not measured.
- **Current rules** Over the mature period the current rules raise 14.0 alerts a day against a capacity of 12. L-02 alone is 50% of the alerts with a precision of 4.7%. 16% of the alerts are open beyond the SLA. Rules with no confirmed fraud: L-06.
- **Fairness** False alerts fall unevenly: the Expat segment is alerted 4.9 times the overall rate by the current rules; with the score the largest gap is Expat at 1.3 times. L-02 takes 88% of its alerts from that segment.
- **Fraud types** Weakest type for the current rules: collusive merchant (0% of its 17 confirmed cases, against 94% for the score). The score finds: account takeover 100%, structuring 100%, card testing 100%, collusive merchant 94%, mule ring 100%, stolen card 76%.

- **Rings** 1 suspected ring(s): 13 accounts on 3 shared devices forwarded USD 460k (69% of what they received), mostly to B-99001, a median 2.8 hours after the money arrived; 3 transaction(s) confirmed so far. The pattern first appears after the training window, so the supervised model could not learn it; the anomaly model and the rules do not need that history.
- **Merchants** 5 suspected merchant(s), led by M-0391 (off-hours trading; round amounts; first digits unusual (Benford); young merchant). Small-amount bursts point to card testing; round amounts at night at a young merchant point to collusion. These are reasons to review, not findings of wrongdoing.
- **Queue** Since the start of the test window the alerts group into 346 cases (8 P1) with an exposure of USD 1.16m, of which USD 1.02m is not confirmed; 427 alerted transactions were seen neither by the current rules nor by an investigator.
- **Economics** On confirmed fraud alone the score is worth about USD 417k a year more than the current rules, net of the review cost and customer friction. The best team size on these numbers is 2 analyst(s).
- **Stability** Score drift (PSI) is 0.044; the largest feature drift is Merchant's reported fraud rate at 0.075. Precision moved from 13.6% in validation to 16.2% in test.
- **Data** 35 transaction(s) excluded and listed, 40 duplicates removed, 3 day(s) of amounts sent in minor units corrected (482 rows), 8 case(s) pointing to no transaction. dispute_flag is filled in after the event and was kept out of every model.
- **Overall** Fraud controls have material gaps: 11 tests passed, 11 with a warning, 3 failed. Priority actions: Clear the aged alerts by risk score; agree who owns the queue. Review the rule behind the gap; compliance decides whether the gap is justified.
- **Analyst** Illustrative note (synthetic data): the head of fraud operations confirms that L-02 was tightened after an incident two years ago and has not been reviewed since.
- **Analyst** Illustrative note (synthetic data): the money-laundering reporting officer has been told about the shared-device ring and will decide whether a report is due.

Actions

Test	Status	Action
Current alerts open beyond the SLA	FAIL	Clear the aged alerts by risk score; agree who owns the queue.
Current rules: largest false-alert ratio by group	FAIL	Review the rule behind the gap; compliance decides whether the gap is justified.
Value through suspected rings, not confirmed	FAIL	Open the ring case now: hold outgoing transfers, review KYC, MLRO decides on reporting.
Cases that point to no transaction	WARN	Reconcile the case system with the transaction system; the orphans are listed.
Days sent in the wrong unit	WARN	Ask the owner of the source system to add a unit control to the feed.
Fields known only after the event	WARN	Keep these fields out of every model; tell the data owner when each one is filled in.
Score alerts against the team's capacity	WARN	Reset the threshold on recent data; high-severity rules are firing more than in the validation window.
Precision of the current rules	WARN	Retire or tighten the noisiest rules (see the current rules table), starting with the largest.
Fraud value found by the current rules	WARN	Add the new rule pack beside the current rules; run them in shadow mode for a month.
Current alerts against the team's capacity	WARN	Cut alert volume before adding people: retire noisy rules first.
Current rules that never confirmed fraud	WARN	Retire the rules with no confirmed fraud after a documented review.
Value at suspected merchants, not confirmed	WARN	Review the suspected merchants and consider a settlement hold.
Alert value the current rules do not see	WARN	Work the new leads in the case review sheet and add the rules that find them.
Share of the confirmed loss recovered	WARN	Review the recovery process: speed of the recall request and the chargeback handling.

Source: tables/tests.csv

Methodology

This report tells what was found. The steps, formulas and parameters are in the methodology document. One thing helps when reading on: every result below comes from a period the models never saw, measured at the number of alerts the team can really review.

Results

1. The picture

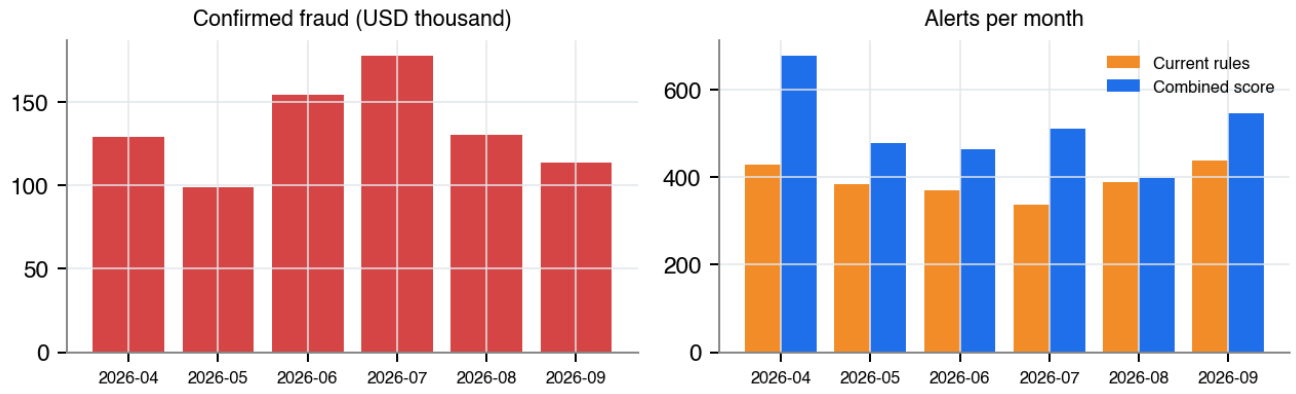


Figure 1. Confirmed fraud and alerts by month. The last month is immature. Source: tables/monthly.csv

Window	From	To	Transactions	Confirmed	Alerts a day	Precision	Value found	PR-AUC
burn_in	2026-04-01	2026-05-16	20,297	96	20.2	9.1%	95%	0.53
train	2026-05-16	2026-07-28	32,685	218	15.5	18.6%	99%	0.55
validation	2026-07-28	2026-08-21	10,900	47	12.0	13.6%	98%	0.47
test	2026-08-21	2026-09-14	11,097	82	18.8	16.2%	99%	0.34
live	2026-09-15	2026-09-30	7,213	23	14.4	7.8%	95%	0.25

Source: tables/windows.csv

2. What each method finds, out of time

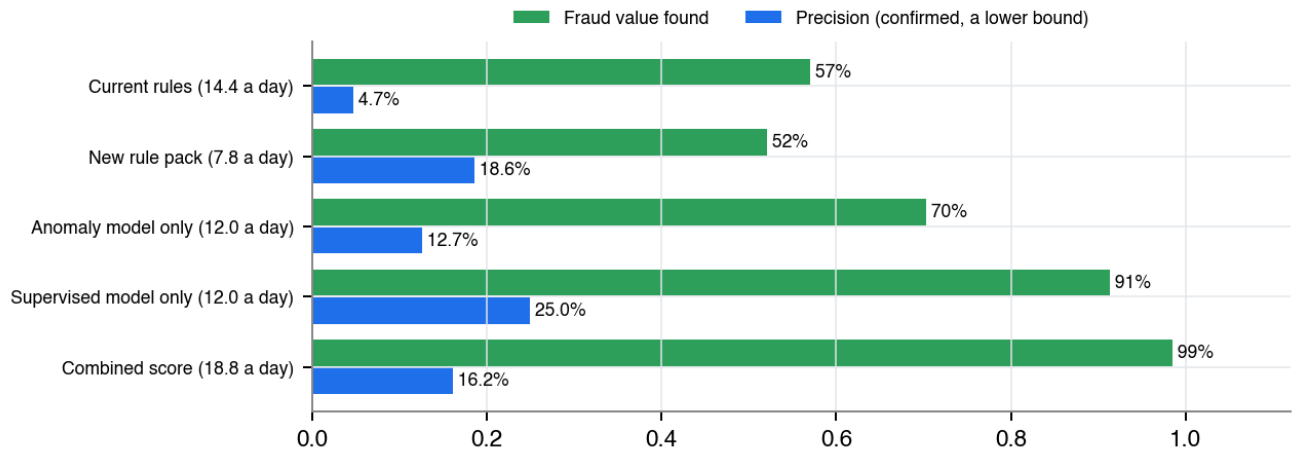


Figure 2. Value found and precision in the test window. Source: tables/method_comparison.csv

Method	Alerts a day	Found	Precision	Low	High	Cases	Value	V. low	V. high
Current rules	14.4	17	4.7%	2.7%	7.1%	21%	57%	36%	71%
New rule pack	7.8	36	18.6%	13.0%	24.6%	44%	52%	32%	68%
Anomaly model only	12.0	38	12.7%	9.0%	16.7%	46%	70%	54%	82%
Supervised model only	12.0	75	25.0%	20.1%	30.2%	91%	91%	80%	99%
Combined score	18.8	76	16.2%	12.8%	19.7%	93%	99%	97%	100%

Source: tables/method_comparison.csv

How to read this. Precision counts what someone has already confirmed. Most of what the score raises has never been looked at, so its real precision can only be higher; the current rules are favoured, because many confirmed cases exist only because those rules fired. The intervals are 95% bootstrap intervals; a difference inside them is not evidence.

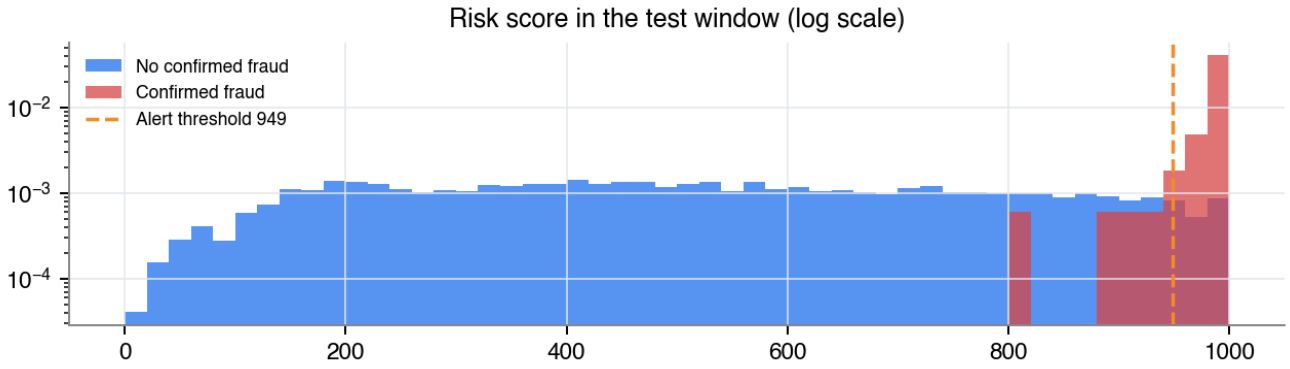


Figure 3. Score of confirmed fraud against the rest. Source: interim scored transactions

3. How many alerts?

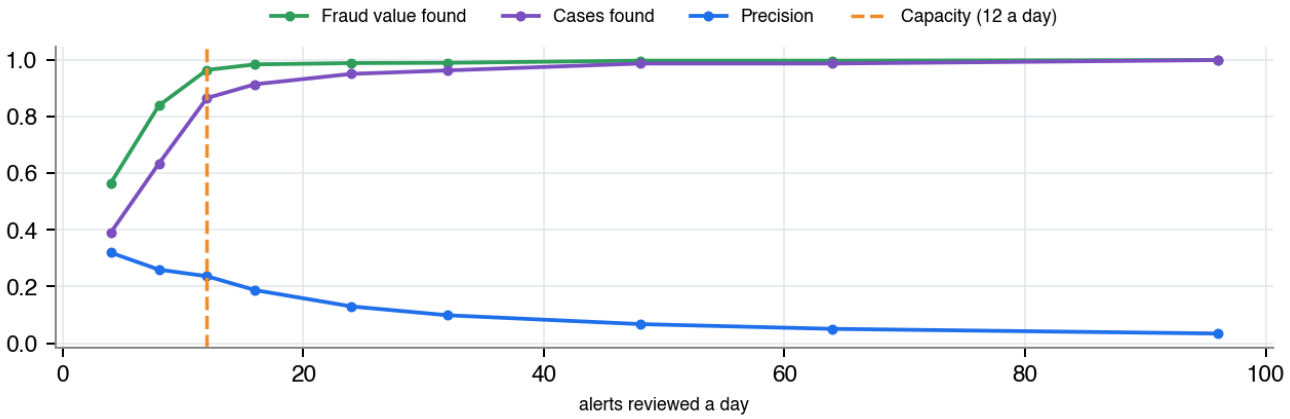


Figure 4. What each extra alert buys, on the test window. Source: tables/threshold_sweep.csv

Alerts a day	Precision	Cases found	Value found	Confirmed	No confirmed fraud
4	32.0%	39%	57%	32	68
8	26.0%	63%	84%	52	148
12	23.7%	87%	97%	71	229
16	18.8%	91%	98%	75	325
24	13.0%	95%	99%	78	522
32	9.9%	96%	99%	79	721
48	6.8%	99%	100%	81	1,119
64	5.1%	99%	100%	81	1,519
96	3.4%	100%	100%	82	2,318

Source: tables/threshold_sweep.csv

4. Fraud types

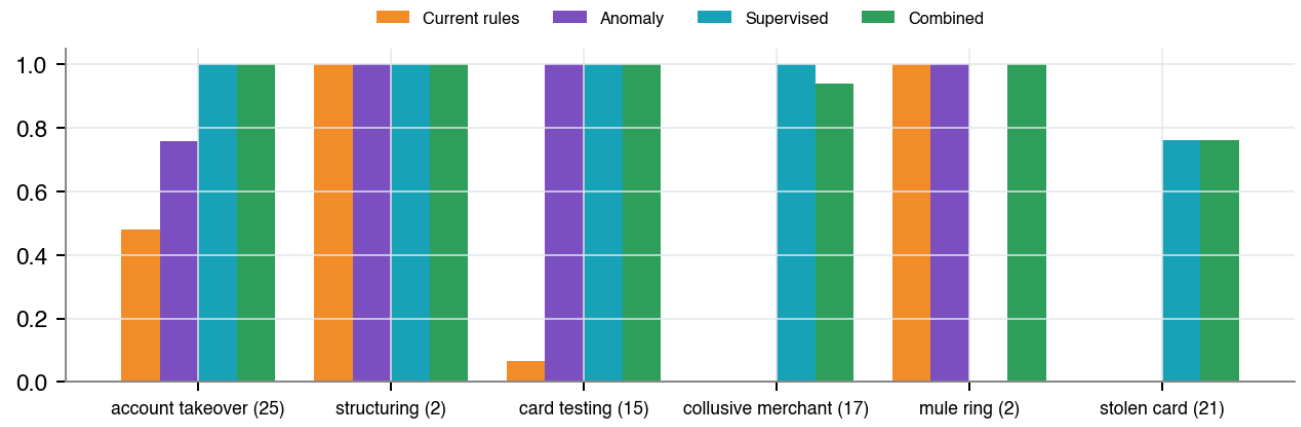


Figure 5. Share of confirmed cases found, by type and method. Source: tables/typology_recall.csv

Type	Confirmed	USD	Current	Rule pack	Anomaly	Supervised	Combined
account takeover	25	48,417	48%	48%	76%	100%	100%
structuring	2	18,748	100%	100%	100%	100%	100%
card testing	15	12,366	7%	60%	100%	100%	100%
collusive merchant	17	8,500	0%	65%	0%	100%	94%
mule ring	2	7,627	100%	100%	100%	0%	100%
stolen card	21	6,542	0%	0%	0%	76%	76%

Source: tables/typology_recall.csv

Types with few confirmed cases are shown as they are: three cases are not a measurement. The anomaly model needs no labels, which is why it matters for patterns nobody has confirmed yet.

5. The current rules

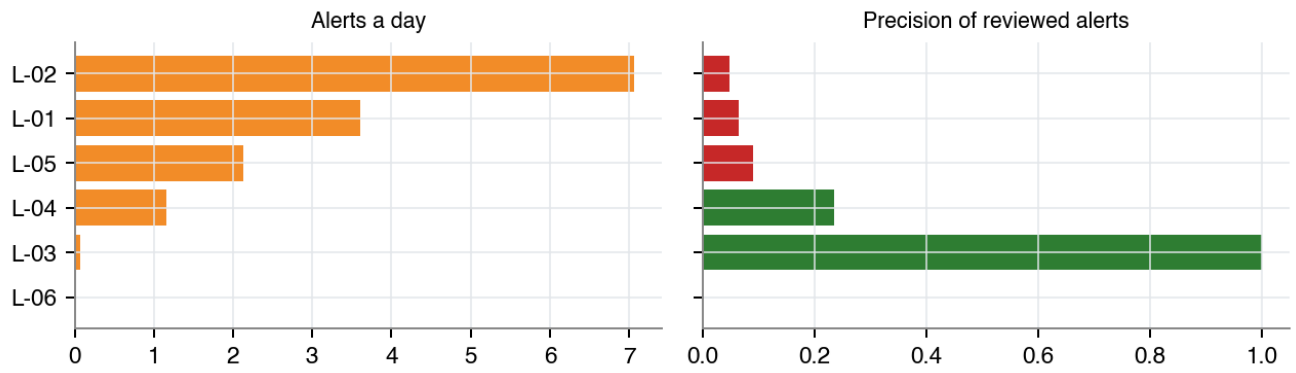


Figure 6. Volume and precision of each current rule. Source: tables/legacy_rules.csv

Rule	Alerts	A day	Reviewed	Aged	Confirmed	Precision	Value found
L-02	1,179	7.1	995	184	47	4.7%	157,904
L-01	602	3.6	488	114	31	6.4%	300,159
L-05	356	2.1	311	45	28	9.0%	80,497
L-04	193	1.2	166	27	39	23.5%	133,017
L-03	12	0.1	10	2	10	100.0%	22,705
L-06	3	0.0	3	0	0	0.0%	0

Source: tables/legacy_rules.csv

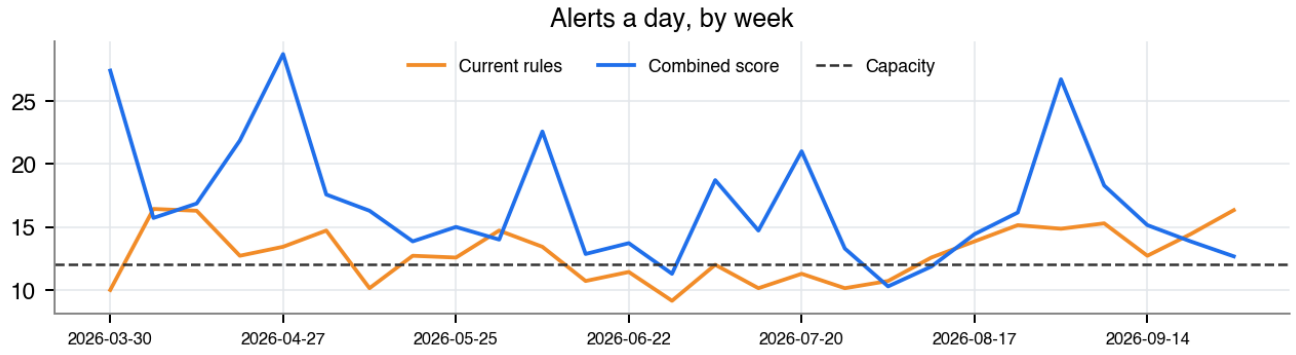


Figure 7. Alerts a day against the capacity. Source: tables/weekly_alerts.csv

6. The new rule pack

Rule	What it looks for	Type	Severity	Hits	Confirmed	Precision	Only this rule
R-01	New device, foreign IP, large for this account	Account takeover	high	3	2	66.7%	0
R-02	Burst of value on a new device	Account takeover	high	12	9	75.0%	7
R-03	New payee, large for this account, on a recently new device	Account takeover	medium	17	5	29.4%	1
R-04	Small-amount burst at a merchant	Card testing	high	40	9	22.5%	9
R-05	Money forwarded within a day from a shared device	Mule activity	high	69	2	2.9%	0
R-06	Young account forwards most of what it received	Mule activity	medium	69	2	2.9%	0
R-07	Repeated amounts just under the reporting threshold	Structuring	high	4	2	50.0%	2
R-08	Round amount at night at a young merchant	Merchant collusion	medium	47	11	23.4%	11
R-09	Impossible travel	Card compromise	medium	1	0	0.0%	0
R-10	Dormant account wakes up with a large payment	Account takeover	low	6	0	0.0%	0

Source: tables/rule_performance.csv

Measured on the test window. A low confirmed precision on a mule rule usually means nobody has confirmed those transactions yet, not that the rule is wrong; see section 8. The pack is delivered in YAML and SQL so that it can run in the client's own system.

7. Who bears the false alerts

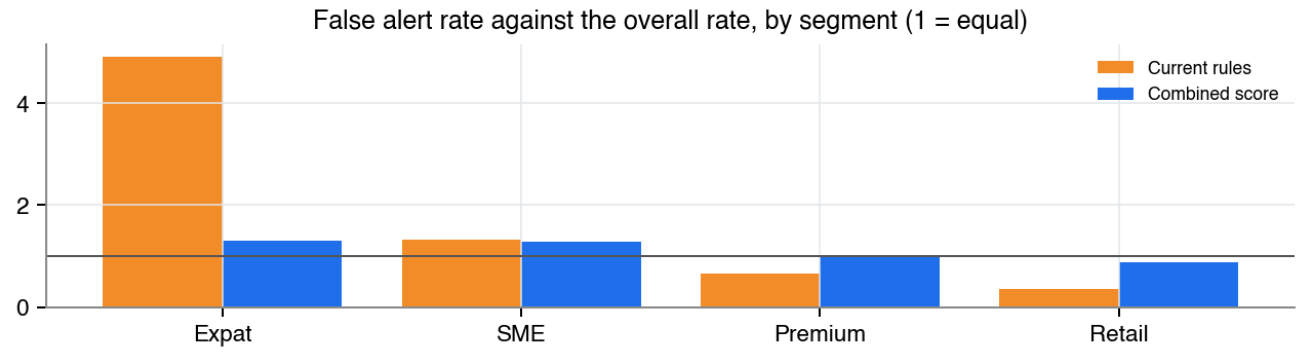


Figure 8. False alert rate by segment. Source: tables/fairness.csv

Dimension	Group	Method	Transactions	False alert rate	Against overall
segment	Expat	Current rules	1,112	15.4%	4.92
segment	SME	Current rules	1,681	4.1%	1.31
segment	Premium	Current rules	1,511	2.1%	0.66
segment	Retail	Current rules	6,711	1.1%	0.35
segment	Expat	Combined score	1,112	4.7%	1.31
segment	SME	Combined score	1,681	4.6%	1.28
segment	Premium	Combined score	1,511	3.6%	1.00
segment	Retail	Combined score	6,711	3.1%	0.88
home_country	DE	Current rules	1,011	4.5%	1.43
home_country	GB	Current rules	1,678	3.6%	1.16
home_country	SA	Current rules	2,378	3.2%	1.01
home_country	AE	Current rules	4,843	2.8%	0.89
home_country	IN	Current rules	1,105	2.5%	0.81
home_country	SA	Combined score	2,378	4.7%	1.30
home_country	DE	Combined score	1,011	4.3%	1.19
home_country	AE	Combined score	4,843	3.3%	0.92
home_country	GB	Combined score	1,678	3.3%	0.92
home_country	IN	Combined score	1,105	2.4%	0.66

Source: tables/fairness.csv

Being alerted is an inconvenience with a cost: a blocked payment, a call, a lost customer. A gap between groups is not automatically wrong, but it has to be understood and justified; compliance decides.

8. Rings and shared devices

Group	Status	Accounts	Devices	Top beneficiary	Out (USD)	Out / in	Hold (h)	Confirmed
RING-071	Suspected ring	13	3	B-99001	459,500	69%	2.8	3
RING-141	Compromised-device pattern	45	1	B-00148	225,557	22%	588.3	11
RING-140	Compromised-device pattern	45	1	B-00985	222,304	22%	629.3	25
RING-139	Compromised-device pattern	45	1	B-01896	156,569	14%	574.1	14
RING-142	Compromised-device pattern	45	1	B-00046	154,224	14%	647.0	15

Source: tables/rings.csv

Groups of three or more accounts on shared devices. A suspected ring forwards most of what it receives within hours; a compromised-device pattern is one device on very many accounts, typical of card testing. Households sharing one device are not shown.

9. Merchants

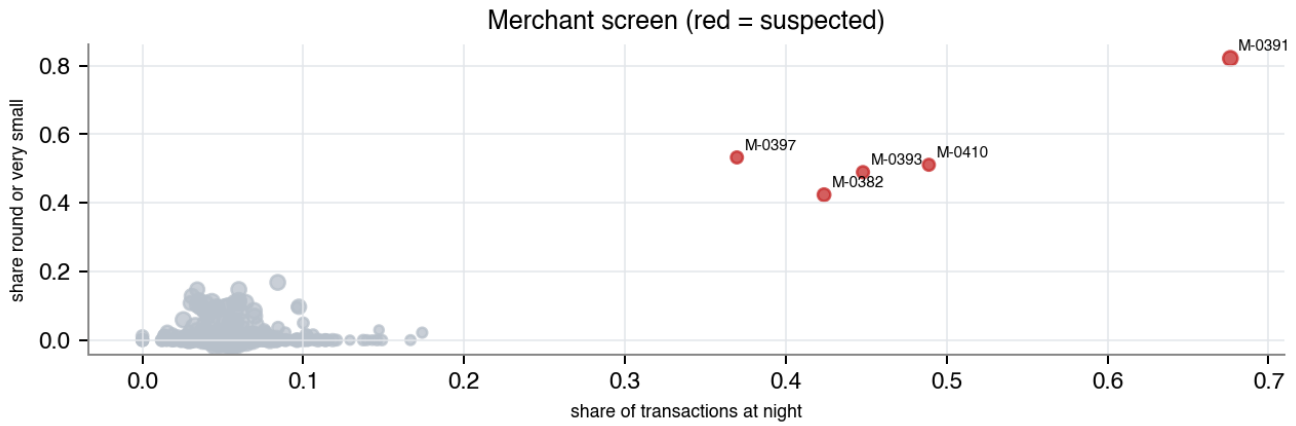


Figure 9. Merchant screen. Source: tables/merchant_screen.csv

Merchant	Status	Txns	USD	Night	Round	Small	Benford	Confirmed	Signals
M-0391	Suspected	207	60,612	68%	82%	0%	0.080	52	off-hours trading; round amounts; first digits unusual (Benford); young merchant
M-0382	Suspected	118	14,799	42%	0%	42%	0.053	15	off-hours trading; small-amount bursts; first digits unusual (Benford)
M-0393	Suspected	96	7,341	45%	0%	49%	0.051	15	off-hours trading; small-amount bursts; first digits unusual (Benford)
M-0397	Suspected	92	6,020	37%	0%	53%	0.052	7	off-hours trading; small-amount bursts; first digits unusual (Benford)
M-0410	Suspected	90	3,502	49%	0%	51%	0.065	10	off-hours trading; small-amount bursts; first digits unusual (Benford)

Source: tables/merchant_screen.csv

10. The case queue

Priority	Type	Cases	Exposure (USD)
P1	Account takeover	3	24,563
P1	Card testing	2	3,376
P1	Merchant collusion	1	26,100
P1	Mule activity	1	409,751
P1	Structuring	1	37,523
P2	Account takeover	8	17,448
P2	Unusual behaviour	108	540,196
P3	Account takeover	9	5,289
P3	Unusual behaviour	213	96,973

Source: tables/case_queue.csv

The fifteen first cases

Case	Priority	Entity	ID	Type	Txns	Exposure	Unconfirmed	Score
Q-0001	P1	Ring	RING-071	Mule activity	138	409,751	397,858	997
Q-0002	P1	Account	A-00592	Structuring	9	37,523	18,776	999
Q-0003	P1	Merchant	M-0391	Merchant collusion	76	26,100	17,400	996
Q-0004	P1	Account	A-00747	Account takeover	6	12,549	822	999
Q-0005	P1	Account	A-01596	Account takeover	2	6,967	6,967	970
Q-0006	P1	Account	A-02375	Account takeover	1	5,047	5,047	969
Q-0007	P1	Merchant	M-0397	Card testing	1	3,297	3,297	996
Q-0008	P1	Merchant	M-0410	Card testing	45	79	62	999
Q-0009	P2	Account	A-00715	Unusual behaviour	3	29,475	29,475	962
Q-0010	P2	Account	A-00817	Unusual behaviour	2	20,313	20,313	974
Q-0011	P2	Account	A-01063	Unusual behaviour	2	19,762	19,762	983
Q-0012	P2	Account	A-00322	Unusual behaviour	8	18,014	0	999
Q-0013	P2	Account	A-00752	Unusual behaviour	5	15,944	2,772	999
Q-0014	P2	Account	A-02509	Unusual behaviour	7	15,939	4,307	999
Q-0015	P2	Account	A-01534	Unusual behaviour	1	15,411	15,411	958

Source: tables/case_queue.csv

The full queue, with the reasons, is in the case review sheet; the first cases are written up in the investigator brief.

11. Economics and team size

Method	Alerts a day	Prevented	Review cost	Friction	Net a year
Current rules	14.4	596,152	47,435	15,067	533,649
New rule pack	7.8	544,651	25,492	6,920	512,239
Anomaly model only	12.0	734,280	39,420	11,476	683,384
Supervised model only	12.0	953,621	39,420	9,855	904,346
Combined score	18.8	1,029,283	61,758	17,257	950,268

Source: tables/economics.csv

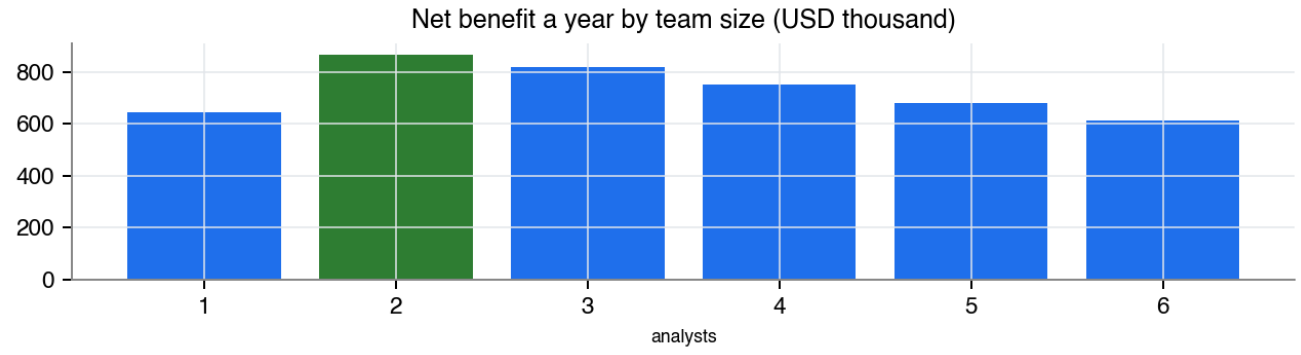


Figure 10. Net benefit by team size. Source: tables/staffing.csv

Analysts	Alerts a day	Precision	Value found	Prevented	Net a year	Last analyst adds
1	6	24.7%	69%	715,890	645,941	n/a
2	12	23.7%	97%	1,008,074	868,043	222,103
3	18	16.9%	99%	1,029,283	817,902	-50,142
4	24	13.0%	99%	1,033,360	750,497	-67,405
5	30	10.5%	99%	1,034,382	679,993	-70,504
6	36	8.9%	100%	1,040,356	614,440	-65,553

Source: tables/staffing.csv

Scaled to a year from the test window, on confirmed fraud only; assumptions are in the configuration and the methodology. These are estimates, not forecasts.

12. What the model relies on, and is it still valid?

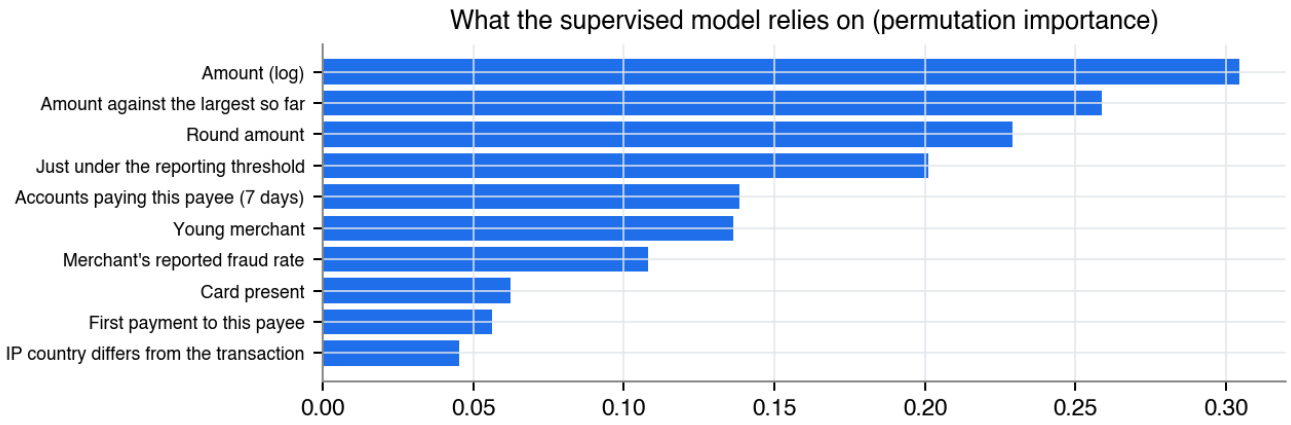


Figure 11. Feature importance. Source: tables/feature_importance.csv

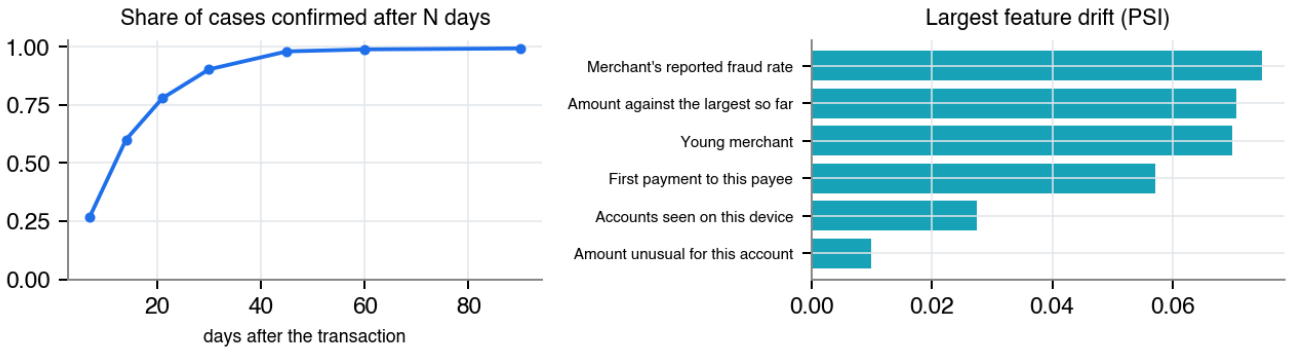


Figure 12. Case delay and drift. Source: tables/label_delay.csv, tables/drift.csv

Leakage screen

Field	Kind	AUC alone	Decision
dispute_flag	client field, not a feature	0.999	Leakage suspected: known only after the event; never used
log_amount	model feature	0.760	Fine
usd_24h	model feature	0.745	Fine
mins_since_last	model feature	0.743	Fine
n_24h	model feature	0.737	Fine
dormant_days	model feature	0.715	Fine

Source: tables/leakage_screen.csv

13. Data quality

100,939 rows were received; 35 were excluded, 40 duplicates removed, 576 refunds kept apart and 482 rows corrected for minor units.

Level	Reason	Count
batch	amounts sent in minor units	3
case	case points to an unknown transaction	8
transaction	dated after the period	5
transaction	unknown account	30

Source: tables/data_exceptions.csv

14. All tests

Area	Test	Result	Threshold	Status
Data	Transactions that can be analysed	99.97%	warn below 99.50%, fail below 98.00%	PASS
Data	Cases that point to no transaction	1.47%	warn above 1.00%, fail above 5.00%	WARN
Data	Days sent in the wrong unit	3 day(s)	warn above 0 day(s), fail above 10 day(s)	WARN
Data	Fields known only after the event	1 field(s)	warn above 0 field(s), fail above 3 field(s)	WARN
Data	Median days to confirm a case	12 days	warn above 14 days, fail above 30 days	PASS
Detection	Precision at the alert budget	16.2%	warn below 15.0%, fail below 5.0%	PASS
Detection	Fraud value found at the alert budget	98.5%	warn below 60.0%, fail below 40.0%	PASS
Detection	Score alerts against the team's capacity	1.57x	warn above 1.10x, fail above 2.00x	WARN
Detection	Precision against the current rules	3.4x	warn below 2.0x, fail below 1.0x	PASS
Detection	Ranking power (PR-AUC against the base rate)	47x	warn below 30x, fail below 10x	PASS
Detection	Precision lost from validation to test	0%	warn above 30%, fail above 60%	PASS
Detection	Weakest fraud type found	stolen_card: 76% of confirmed cases	warn below 60%, fail below 30%	PASS
Current controls	Precision of the current rules	4.7%	warn below 10.0%, fail below 3.0%	WARN
Current controls	Fraud value found by the current rules	57.1%	warn below 60.0%, fail below 30.0%	WARN
Current controls	Current alerts against the team's capacity	1.17x	warn above 1.00x, fail above 1.25x	WARN
Current controls	Current alerts open beyond the SLA	15.9%	warn above 2.0%, fail above 10.0%	FAIL
Current controls	Current rules that never confirmed fraud	17%	warn above 0%, fail above 30%	WARN
Fairness	Current rules: largest false-alert ratio by group	Expat: 4.9x the overall rate	warn above 1.5x, fail above 2.5x	FAIL
Fairness	New score: largest false-alert ratio by group	Expat: 1.3x the overall rate	warn above 1.5x, fail above 2.5x	PASS
Stability	Score drift (PSI)	0.044	warn above 0.100, fail above 0.250	PASS
Stability	Largest feature drift (PSI)	Merchant's reported fraud rate: 0.075	warn above 0.10, fail above 0.25	PASS
Exposure	Value through suspected rings, not confirmed	397,858 USD	warn above 0 USD, fail above 100,000 USD	FAIL
Exposure	Value at suspected merchants, not confirmed	20,759 USD	warn above 0 USD, fail above 100,000 USD	WARN
Exposure	Alert value the current rules do not see	23%	warn above 10%, fail above 30%	WARN
Exposure	Share of the confirmed loss recovered	20.7%	warn below 40.0%, fail below 15.0%	WARN

Source: tables/tests.csv

Limitations and assumptions

- Confirmed cases are a late, partial and biased sample of the fraud. Measured precision and recall are estimates against what is known, not against the truth.
- Selection bias: cases come mostly from customers and from the current rules. Patterns nobody looks for stay unconfirmed, so the supervised model cannot learn them.
- The comparison with the current rules favours them, for the same reason.
- The test window is short; fraud comes in bursts, and intervals are wide. One window is not a guarantee for the next.
- An alert, a suspected ring and a suspected merchant are reasons to review. Nothing in this report is a finding of wrongdoing against any person or business; reporting decisions belong to the client's compliance officer.
- The economics rest on stated assumptions (share stopped, cost of a review, friction).
- The score has not been run in production; a shadow period is recommended before it blocks any payment.

Appendix

Cleaning log

step	action	rows_affected
duplicates	exact duplicate transaction rows removed (processor retry)	40
harmonise	timestamps read from dd/mm/yyyy as well as ISO	41729
harmonise	currency spellings harmonised (usd, US$, spaces)	500
units	days sent in minor units (cents) divided by 100	482
fx	missing FX rates filled from the previous day	2
scope	refunds (negative amounts) kept apart: not scored	576
exclude	transaction excluded: unknown account	30
exclude	transaction excluded: dated after the period	5
cases	outcome spellings harmonised	96
cases	cases entered twice removed	12
cases	cases pointing to an unknown or excluded transaction listed and dropped	8
cases	transactions whose outcome changed: the latest entry counts	3

Source: tables/cleaning_log.csv

Data quality

accounts · 3000 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

alerts · 2592 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	closed_date	406	406 missing (15.66%)

Source: data_quality/dq_alerts.json

cases · 546 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
missing_values	info	fraud_type	65	65 missing (11.90%)
duplicate_rows	warning		12	12 fully duplicated rows
duplicate_keys	warning		24	24 rows share a key ['case_id'] (0 with conflicting values)

Source: data_quality/dq_cases.json

fx_rates · 734 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

merchants · 420 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

transactions · 100939 rows · PASSED (0 errors, 2 warnings)

Check	Severity	Column	Count	Message
missing_values	info	device_id	59845	59845 missing (59.29%)
duplicate_rows	warning		40	40 fully duplicated rows
duplicate_keys	warning		80	80 rows share a key ['txn_id'] (0 with conflicting values)

Source: data_quality/dq_transactions.json

transactions_clean · 100864 rows · PASSED (0 errors, 0 warnings)

Check	Severity	Column	Count	Message
missing_values	info	device_id	59809	59809 missing (59.30%)

Source: data_quality/dq_transactions_clean.json

Audit trail

Run ID	`fa_20261111T142159Z_531350`
Run started / finished (UTC)	2026-11-11T14:21:59+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`3a6bb15275c5 (uncommitted changes) · core 0.1.0`
Config SHA-256	`8191c3fc9de383b6`
Input: transactions	`100939 rows · sha256 59c79c357ce5872c`
Input: accounts	`3000 rows · sha256 9e6df748cb753a9c`
Input: merchants	`420 rows · sha256 43b8424681529e28`
Input: fx_rates	`734 rows · sha256 0d53d2d5aba20d07`
Input: alerts	`2592 rows · sha256 a5d41ff79cd234e5`
Input: cases	`546 rows · sha256 76d5257c1660ec05`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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